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Recliner handle with overload protection device

Abstract

A recliner handle with an overload protection device is provided for a vehicle seat having a recliner attached to a recliner B-bracket. The recliner handle is attached to a handle spline having a disc shaft aperture configured to engage with a disc shaft of the recliner. A stop flange attached to the handle spline is configured to frictionally engage with an overload stop attached to the recliner B-bracket. The handle spline is rotatable in a first rotational direction between a first position wherein the stop flange is frictionally engaged with the overload stop and the disc shaft aperture is disengaged from the disc shaft and a second position wherein the disc shaft aperture engages the disc shaft. Rotation of the handle spline in a second rotational direction from the first position is restricted by the stop flange frictionally engaged with the overload stop.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to and all the benefits of U.S. Provisional Application 63/011,604, filed Apr. 17, 2020, and entitled “Recliner Handle With Overload Protection Device”, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The present invention relates to a recliner handle configured to unlock a recliner of a vehicle seat so that the vehicle seat can be reclined. More specifically, the present invention relates to an overload protection device preventing abuse torque applied to the recliner handle from being transferred to the recliner of the vehicle seat.
2. Description of Related Art
(2) Many vehicles today have vehicle seats with a seat back that is rotatably coupled to a seat

cushion by a recliner. Typical recliners have a locked condition wherein the seat back is locked in a selected rotational position with respect to the seat cushion. Further, the typical recliners have an unlocked condition wherein the seat back can be reclined. One known recliner has a disc shaft projecting from the recliner. Rotating the disc shaft between a home angular position and a release angular position reconfigures the known recliner between the locked condition and the unlocked condition.
(3) A recliner handle is typically operatively coupled to the disc shaft of the known recliner for rotating the disc shaft between the home angular position and the release angular position. Typically, the recliner handle has a home position that corresponds to the disc shaft being in the home angular position with the known recliner being in the locked condition. In addition, the recliner handle typically has a recline release position that corresponds to the disc shaft being in the release angular position with the known recliner being in the unlocked condition. When the recliner handle is in the home position, lifting upward on the recliner handle applies torque to the recliner handle. The upward torque applied to the recliner handle rotates the recliner handle and the connected disc shaft. When the recliner handle reaches the recline release position, the disc shaft is correspondingly rotated to the release angular position, and the known recliner is unlocked.
(4) The known recliner includes a return spring biasing the recliner handle towards the home position to ensure the known recliner is placed in the locked condition when upward torque is not applied to the recliner handle. Thus, removal of upward torque on the recliner handle results in the recliner handle being rotated back to the home position which also rotates the disc shaft back to the home angular position and returns the known recliner to the locked condition.
(5) However, the known recliner can be overloaded and potentially damaged when the recliner handle is in the home position and the recliner handle is pushed downward. In this case, downward torque applied to the recliner handle transfers excessive or abuse torque to the known recliner which can damage internal components within the known recliner.
(6) Thus, it is desirable to prevent damage to the recliner when excessive downward pressure (abuse torque) is applied to the recliner handle when the recliner handle is in the home position. Further, it is desirable to transfer abuse torque applied to the recliner handle away from the recliner. Finally, it is desirable to include an overload protection device between the recliner handle and the recliner configured to avoid overloading the recliner when downward torque is applied to the recliner handle.
SUMMARY OF THE INVENTION
(7) The present invention relates to a recliner handle with an overload protection device for a vehicle seat having a recliner attached to a recliner B-bracket. The recliner handle is fixedly coupled to a handle spline having a disc shaft aperture configured to engage with a disc shaft operatively coupled to the recliner. A stop flange attached to the handle spline is configured to frictionally engage with an overload stop attached to the recliner B-bracket. The handle spline is rotatable in a first rotational direction between a first position wherein the stop flange is frictionally engaged with the overload stop and the disc shaft aperture is disengaged from the disc shaft and a second position wherein the disc shaft aperture engages the disc shaft. Rotation of the handle spline in a second rotational direction from the first position is restricted by the stop flange frictionally engaged with the overload stop.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Advantages of the present invention will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:

(2) FIG. **1** is a fragmentary perspective view of a vehicle seat having an overload protection device operatively coupled between a recliner and a recliner handle, according to one embodiment of the present invention;
(3) FIG. **2** is an exploded view of the overload protection device of FIG. **1**, showing the recliner, a recliner B-bracket having a cutout, a spring bracket, a clinch ring, a handle spline, and a handle inertia spring, according to one embodiment of the present invention;
(4) FIG. **3** is a perspective view of the recliner of FIG. **2**, illustrating a disc shaft projecting from the recliner;
(5) FIGS. **4**A and **4**B are perspective views of the spring bracket of FIG. **2**;
(6) FIG. **5** is a perspective view of the clinch ring of FIG. **2**;
(7) FIG. **6**A is a partially transparent perspective view of the handle spline of FIG. **2**, illustrating a disc shaft aperture within the handle spline;
(8) FIG. **6**B is a perspective view of the handle spline of FIG. **6**A;
(9) FIG. **7** is an enlarged cutaway perspective view of the handle spline and the disc shaft taken along section line **7-7** of FIG. **1**, showing the disc shaft aperture in the handle spline and the disc shaft in a neutral position with the disc shaft aperture disengaged from the disc shaft;
(10) FIG. **8** is an enlarged cutaway perspective view of the handle spline and the disc shaft of FIG. **7**, showing the handle spline rotated in a clockwise direction with respect to the disc shaft with the disc shaft aperture engaged with the disc shaft;
(11) FIG. **9** is an enlarged cutaway perspective view of the handle spline and the disc shaft of FIG. **7**, showing the handle spline rotated in a counterclockwise direction with respect to the disc shaft with the disc shaft aperture engaged with the disc shaft;
(12) FIG. **10** is an exploded perspective view of the recliner of FIG. **1**, illustrating assembly of the spring bracket and the handle spline with the disc shaft of the recliner;
(13) FIG. **11** is a perspective view of the recliner of FIG. **10** after assembly of the spring bracket and handle spline, illustrating insertion of the clinch ring into the handle spline;
(14) FIG. **12** is a cross-sectional view of the recliner and the overload protection device taken along section line **12-12** of FIG. **1**, illustrating the handle spline, the spring bracket, and the clinch ring assembled with the disc shaft;
(15) FIG. **13** is a perspective view of the recliner and the overload protection device of FIG. **11** after assembly of the clinch ring with the handle spline, illustrating assembly of the handle inertia spring with the spring bracket and the recliner B-bracket;
(16) FIG. **14** is a perspective view of the recliner and overload protection device of FIG. **13** in a home angular position with a stop flange of the spring bracket frictionally engaged with a cutout attached to the recliner B-bracket, illustrating the handle spline being rotated towards a recline release position;
(17) FIG. **15** is an enlarged cutaway perspective view of the recliner and the overload protection device of FIG. **14** in the home angular position, illustrating the handle spline being rotated towards the release angular position;
(18) FIG. **16** is a perspective view of the recliner and the overload protection device of FIG. **14** after the handle spline is rotated to the release angular position, illustrating the stop flange of the spring bracket being spaced apart from the cutout of the recliner B-bracket;
(19) FIG. **17** is a perspective view of the recliner and the overload protection device of FIG. **16** after the handle spline is rotated towards the home angular position, illustrating the stop flange of the spring bracket frictionally engaged with the cutout of the recliner B-bracket;
(20) FIG. **18** is a cutaway view of the recliner and the overload protection device of FIG. **17** in the home angular position, illustrating the handle spline and the disc shaft in the neutral position with the disc shaft aperture of the handle spline disengaged from the disc shaft;
(21) FIG. **19** is a cutaway view of the recliner and the overload protection device of FIG. **18** in the home angular position, illustrating abuse torque being applied to the handle spline while the disc

shaft aperture is retained disengaged from the disc shaft;
(22) FIG. **20** is a perspective view of the recliner and the overload protection device of FIG. **19**, illustrating abuse torque applied the handle spline resulting in load being applied to the cutout of the recliner B-bracket;
(23) FIG. **21** is a fragmentary perspective view of a vehicle seat having an overload protection device operatively coupled between a recliner and a recliner handle, according to a second embodiment of the present invention;
(24) FIG. **22** is an exploded view of the overload protection device of FIG. **21**, showing the recliner, a recliner B-bracket having a cutout with an overload stop, a handle spline with an integrated stop flange, and a handle inertia spring, according to a second embodiment of the present invention;
(25) FIG. **23** is a cutaway perspective view of the overload protection device of FIG. **22**, showing the integrated stop flange of the handle spline frictionally engaged with the overload stop on the recliner B-bracket cutout;
(26) FIG. **24** is a partially transparent perspective view of the handle spline of FIG. **23**, illustrating a disc shaft aperture passing through the handle spline;
(27) FIG. **25** is a perspective view of the handle spline of FIG. **24**, illustrating exterior surfaces of the handle spline; and
(28) FIG. **26** is an enlarged cutaway perspective view of the handle spline and the disc shaft of FIG. **23**, showing the disc shaft aperture in the handle spline and the disc shaft in a neutral position with the disc shaft aperture disengaged from the disc shaft.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(29) FIGS. **1** through **26** illustrate an overload protection device **10**, **10**′ connecting a recliner handle **14** to a recliner **18** of a vehicle seat **22** according to embodiments described herein. Directional references employed or shown in the description, figures or claims, such as top, bottom, upper, lower, upward, downward, lengthwise, widthwise, left, right, and the like, are relative terms employed for ease of description and are not intended to limit the scope of the invention in any respect. Referring to the Figures, like numerals indicate like or corresponding parts throughout the several views.
(30) FIG. **1** shows a perspective view of a portion of the vehicle seat **22** having a seat back **26** rotatably connected to a seat cushion **30** by the recliner **18**, according to a first embodiment. The recliner handle **14** is fixedly coupled to the overload protection device **10** and rotatably coupled to the recliner **18**. As shown in FIG. **1**, the recliner handle **14** is repositionable between a home position **14**A and a recline release position **14**B about an axis of rotation **34** of the recliner **18**. The home position **14**A of the recliner handle **14** corresponds to a home angular position **36**A of the recliner **18**. Further, the recline release position **14**B of the recliner handle **14** corresponds to a release angular position **36**B of the recliner **18**.
(31) Applying upward torque on the recliner handle **14** and rotating the recliner handle **14** in a first rotational direction A (illustrated by arrow A shown in FIG. **1**) about the axis of rotation **34** of the recliner **18** from the home position **14**A towards the recline release position **14**B unlocks the recliner **18** such that the seat back **26** can be rotated with respect to the seat cushion **30**. In the embodiment shown in FIG. **1**, the recliner handle **14** is rotated upward in a clockwise direction A with respect to the axis of rotation **34** of the recliner **18** to unlock the recliner **18**.
(32) Referring to FIG. **11**, a return spring **38** is operatively coupled between the recliner **18** and the recliner handle **14**. As shown in FIG. **1**, the return spring **38** rotationally biases the recliner handle **14** towards the home position **14**A. When the recliner handle **14** is spaced apart from the home position **14**A, the return spring **38** automatically rotates the recliner handle **14** in a second rotational direction B (arrow B) towards the home position **14**A when the recliner handle **14** is released. The second rotational direction B is opposite the first rotational direction A. If the recliner handle **14** is rotated upward in the clockwise direction A to reposition the recliner handle **14** from the home

position **14**A to the recline release position **14**B, then the recliner handle **14** is rotated downward in the counterclockwise direction B to reposition the recliner handle **14** from the recline release position **14**B back to the home position **14**A.

(33) With certain recliners **18**, further rotation of the recliner handle **14** in the second rotational direction B past the home position **14**A (illustrated by arrow C shown in FIG. **1**) can overload the recliner **18** and cause damage to internal components of the recliner **18**. In addition, rotational motion of the recliner handle **14** in the direction of arrow C past the home position **14**A can overlock the recliner **18**. Overlocking the recliner **18** can potentially damage the recliner **18**. The overload protection device **10** prevents the recliner **18** from being overlocked when downward torque is applied to the recliner handle **14** when the recliner handle **14** is in the home position **14**A. More specifically, the overload protection device **10** restricts rotation of the recliner handle **14** past the home position **14**A in the second rotational direction C and diverts applied torque to the recliner handle **14** away from the recliner **18**.

(34) An exploded view of the overload protection device **10** is shown in FIG. **2**. Referring to FIG. **2**, the recliner **18** includes a disc recliner assembly **42** having a guide plate **42**A rotatably coupled to a tooth plate **42**B. An exemplary disc recliner assembly **42** is an iDiSC 5 disc recliner assembly manufactured by Magna Seating Inc. As shown in FIG. **1**, the guide plate **42**A of the disc recliner assembly **42** is fixedly coupled to a recliner B-bracket **44** with the recliner B-bracket **44** being fixedly coupled to the seat cushion **30**. The guide plate **42**A includes a plurality of bosses **46**, **48** configured to matingly engage with respective notches **50**, **52** in the recliner B-bracket **44**. It is understood that in alternate embodiments, attachment of the disc recliner assembly **42** to the recliner B-bracket **44** may include different means of fixedly coupling the guide plate **42**A to the recliner B-bracket **44** without varying the scope of the invention.

(35) As best shown in FIG. **10**, the tooth plate **42**B is fixedly coupled to the seat back **26**. Referring to FIGS. **1** and **10**, when the tooth plate **42**B is unlocked with respect to the guide plate **42**A (by rotating the recliner handle **14** to the recline release position **14**B), the seat back **26** is rotatable with respect to the recliner B-bracket **44** and the attached seat cushion **30**. In certain embodiments, the recliner B-bracket **44** and the attached seat cushion **30** are also rotatable with respect to the seat back **26** when the guide plate and tooth plates **42**A, **42**B are unlocked relative to each other.

(36) Referring to FIGS. **1** and **10**, the tooth plate **42**B is locked to the guide plate **42**A when the recliner handle **14** is in the home position **14**A. Rotation of the tooth plate **42**B with respect to the guide plate **42**A is prevented while the recliner handle **14** is in the home position **14**A. Thus, rotation of the seat back **26** with respect to the seat cushion **30** is prevented when the recliner handle **14** is in the home position **14**A.

(37) Referring to FIG. **2**, a disc shaft **62** projects from the disc recliner assembly **42** with a longitudinal axis **34**A of the disc shaft **62** aligned with the axis of rotation **34** of the recliner **18**. The disc shaft **62** is formed of a metal, a plastic, and/or combinations thereof. Further, the disc shaft **62** is configured to lock and unlock the rotation of the tooth plate **42**B with respect to the guide plate **42**A based on the angular position **36**A, **36**B of the disc shaft **62**. Rotating the disc shaft **62** from the home angular position **36**A to the release angular position **36**B in first rotational direction A unlocks the guide plate **42**A and the tooth plate **42**B such that one of the guide plate **42**A and the tooth plate **42**B can be rotated with respect to the other one of the guide plate **42**A and the tooth plate **42**B. In other embodiments, both the guide plate **42**A and the tooth plate **42**B can be rotated about the axis of rotation **34** when the disc recliner assembly **42** is unlocked. Thus, in certain embodiments, the seat cushion **30** can be rotated towards and/or away from the seat back **26** as well as the seat back **26** being rotatable towards and/or away from the seat cushion **30**.

(38) An enlarged view of the recliner **18** is shown in FIG. **3**. Referring to FIGS. **2** and **3**, the disc shaft **62** includes an elongated shaft end portion **66** extending along the longitudinal axis **34**A of the disc shaft **62**. The elongated shaft end portion **66** has a rounded rectangular shaped cross-section with opposing shaft flat side portions **72**, **74**, shaft curved side portions **78**, **78**′ extending

between the adjacent shaft flat side portions **72**, **74**, and a distal end surface **80**. The distal end surface **80** of the elongated shaft end portion **66** of the disc shaft **62** is also shown in FIGS. **7**, **8**, and **9**.

(39) As shown in FIGS. **2** and **3**, the disc shaft **62** has a base portion **82** having a generally cylindrical cross-sectional shape with a plurality of flat sides **86**, **90**, **90′** and a base end face **94**. A cross-sectional view of a portion of the overload protection device **10** is shown in FIG. **12** and illustrates additional details of the disc shaft **62**. Referring to FIG. **12**, a central cylindrical portion **98** projects from the base end face **94** of the base portion **82** along the longitudinal axis **34**A of the disc shaft **62**. The central cylindrical portion **98** has an outer diameter **98**A that is less than a maximum diameter **82**A of the base portion **82**. Further, the central cylindrical portion **98** has a central end face **102** extending generally parallel to and spaced apart from the base end face **94** of the base portion **82**. A channel **106** extends circumferentially around an outer perimeter **98′** of the central cylindrical portion **98** spaced apart from both the central end face **102** and the base end face **94**. The channel **106** has an outer diameter **106**A that is less than the outer diameter **98**A of the central cylindrical portion **98**. The elongated shaft end portion **66** projects from the central end face **102** of the central cylindrical portion **98**. Further, the elongated shaft end portion **66** has a maximum diameter **66**A that is less than the outer diameter **98**A of the central cylindrical portion **98**. Also, the base portion **82**, the central cylindrical portion **98**, and the elongated shaft end portion **66** are aligned with the longitudinal axis **34**A of the disc shaft **62**. It will be appreciated that the size, shape, and length of the disc shaft **62** may vary without altering the scope of the invention.

(40) Referring to FIGS. **2**, **3**, and **11**, the return spring **38** is a spiral torsional spring having a first spring end **38**A fixedly coupled to the base portion **82** of the disc shaft **62** and a second spring end **38**B fixedly coupled to a slot **142** in a boss **146** projecting from the guide plate **42**A. The first spring end **38**A of the return spring **38** is wrapped around the base portion **82** of the disc shaft **62** such that the first spring end **38**A frictionally engages with one or more of the flat sides **86**, **90**, **90′** of the base portion **82**. In certain embodiments, the return spring **38** is sized and shaped such that spring tension retains the first spring end **38**A in frictional engagement with the base portion **82** of the disc shaft **62** and retains the second spring end **38**B in an engaged position with the slot **142** in the boss **146** projecting from the guide plate **42**A. In alternate embodiments, one or both of the first and second spring ends **38**A, **38**B are fixedly coupled to the disc shaft **62** and the slot **142**, respectively, by a mechanical fastener, a welded connection, or the like. With the second spring end **38**B of the return spring **38** retained within the slot **142** in the boss **146** projecting from the guide plate **42**A, rotating the recliner handle **14** upward from the home position **14**A towards the recline release position **14**B rotates the disc shaft **62** from the home angular position **36**A to the release angular position **36**B and preloads the return spring **38**. When the recliner handle **14** is released, the spring torque in the return spring **38** urges the disc shaft **62** to rotate towards the home angular position **36**A. Returning the disc shaft **62** to the home angular position **36**A returns the recliner handle **14** to the home position **14**A and locks the disc recliner assembly **42**. It will be appreciated that the size and shape of the return spring **38** may vary without altering the scope of the invention.

(41) As shown in FIG. **2**, the overload protection device **10** includes the disc shaft **62** projecting from the recliner **18**, a spring bracket **158**, a clinch ring **162**, a handle spline **166**, a handle inertia spring **170**, and a tab **174** formed on the recliner B-bracket **44**. It will be appreciated that the overload protection device **10** may include additional components without varying the scope of the invention.

(42) Referring to FIGS. **2**, **4**A, and **4**B, the spring bracket **158** has a generally flat main bracket portion **186** with opposing upper and lower bracket surfaces **190**, **194** spaced apart by first and second bracket side walls **198**, **202** defining a width of the spring bracket **158**. Extending from the main bracket portion **186** are a first flange **206**, a second flange **207**, and a stop flange **208**. The first flange **206** extends upward from a first end **210** of the main bracket portion **186** such that the first flange **206** and the upper bracket surface **190** of the main bracket portion **186** are generally

perpendicular to one another and form an "L" shape. The first flange **206** has opposing inner and outer surfaces **206**A, **206**B and an elongated spring slot **214** extending between the opposing inner and outer surfaces **206**A, **206**B. The second flange **207** extends upward from a second end **222** of the main bracket portion **186**. The second flange **207** includes a flange base portion **226** adjacent the main bracket portion **186** and a distal flange end portion **230** that projects at an angle from the flange base portion **226**. The flange base portion **226** is generally parallel to the first flange **206** with the distal flange end portion **230** being generally parallel to the main bracket portion **186**. The stop flange **208** extends downward from the second bracket side wall **202** of the main bracket portion **186**. The stop flange **208** includes distal and proximal side walls **238**, **242**. The proximal side wall **242** includes a stop surface **246**.

(43) As shown in FIGS. **2**, **4**A, and **4**B, an alignment aperture **250** extends between the upper and lower bracket surfaces **190**, **194** of the main bracket portion **186**. The alignment aperture **250** has a contoured inner profile **250**′ with a plurality of splines **254** extending between the opposing upper and lower bracket surfaces **190**, **194**. The plurality of splines **254** includes at least one narrow spline **254**A and at least one wide spline **254**B configured to matingly engage with at least one narrow spline **286**A and at least one wide spline **286**B located on the handle spline **166**. The wide spline **254**B of the alignment aperture **250** is an alignment feature for assembling the spring bracket **158** and the handle spline **166** with a predetermined relative orientation. Further, the alignment aperture **250** has a minimum opening diameter **250**A and a maximum opening diameter **250**B. The minimum opening diameter **250**A of the alignment aperture **250** is less than the maximum diameter **82**A of the base portion **82** of the disc shaft **62**, as shown in FIG. **12**. The spring bracket **158** is assembled with the disc shaft **62** with the disc shaft **62** passing through the alignment aperture **250**. After assembly, the lower bracket surface **194** of the spring bracket **158** is supported by the base end face **94** of the base portion **82** of the disc shaft **62**. In addition, the spring bracket **158** is formed of a metal, a plastic, and/or combinations thereof. It will be appreciated that the size and shape of the spring bracket **158**, including the first and second flanges **206**, **207**, the stop flange **208**, the spring slot **214**, and the alignment aperture **250** may vary without altering the scope of the invention.

(44) As shown in FIG. **5**, the clinch ring **162** is a generally ring-shaped disc with opposing top and bottom surfaces **162**A, **162**B, an outer side surface **162**C extending between the opposing top and bottom surfaces **162**A, **162**B, and a passageway **258** extending between the opposing top and bottom surfaces **162**A, **162**B forming a notch **258**′ in the clinch ring **162**. The notch **258**′ defining the passageway **258** includes a first lead-in surface **266**, a second lead-in surface **266**′, and an inner cylindrical surface **262** extending between inner ends **266**A, **266**B of the adjacent first and second lead-in surfaces **266**, **266**′. The inner cylindrical surface **262** has a ring inner radius **262**A approximately equal to or greater than half of the outer diameter **106**A of the channel **106** in the central cylindrical portion **98** of the disc shaft **62**. The first and second lead-in surfaces **266**, **266**′ are configured to allow attachment of the clinch ring **162** to the disc shaft **62** within the channel **106** in the central cylindrical portion **98** of the disc shaft **62**. The clinch ring **162** is formed of a metal, a plastic, and/or combinations thereof. It will be appreciated that the size and shape of the clinch ring **162** may vary without altering the scope of the invention.

(45) As shown in FIGS. **2**, **6**A, and **6**B, the handle spline **166** has a splined portion **270** having a generally cylindrical shape with an outer surface **274** extending between an upper end surface **278** and a lower end surface **282**. The outer surface **274** of the handle spline **166** has an outer profile **274**′ sized and shaped to matingly engage with the inner profile **250**′ of the alignment aperture **250** in the spring bracket **158**. Referring to FIGS. **6**A and **6**B, the outer surface **274** of the handle spline **166** includes a plurality of splines **286** spaced around the outer profile **274**′ of the splined portion **270** and extending along a longitudinal axis **34**B of the handle spline **166**. The longitudinal axis **34**B of the handle spline **166** is aligned with the axis of rotation **34** of recliner **18** and the longitudinal axis **34**A of the disc shaft **62** when the handle spline **166** is assembled with the disc

shaft **62**, as illustrated in FIG. **2**.
(46) Referring to FIGS. **4**A and **6**B, the plurality of splines **286** of the handle spline **166** includes at least one narrow spline **286**A and at least one wide spline **286**B configured to matingly engage with the at least one narrow spline **254**A and the at least one wide spline **254**B within the alignment aperture **250** of the spring bracket **158** when the handle spline **166** is assembled with the spring bracket **158**. The wide splines **254**B, **286**B of the alignment aperture **250** and the handle spline **166** are alignment features for assembling the spring bracket **158** and the handle spline **166** with a predetermined relative orientation.
(47) The assembly of the handle spline **166** and the spring bracket **158** onto the disc shaft **62** is shown in FIG. **10**. FIG. **12** shows a cross-sectional view of the handle spline **166** assembled with the spring bracket **158** with the lower end surface **282** of the handle spline **166** inserted into the alignment aperture **250** in the spring bracket **158**.
(48) As shown in FIG. **6**A, the splined portion **270** of the handle spline **166** optionally includes a handle channel **288** extending around the outer periphery of the splined portion **270** and spaced apart from the upper end surface **278** of the splined portion **270**. The handle channel **288** is sized and shaped to matingly engage with and retain a snap feature on the recliner handle **14**. In other embodiments, the handle channel **288** is sized and shaped to receive a clip ring or similar fastening device to retain the recliner handle **14** in an engaged position with the handle spline **166**. In certain embodiments, the handle channel **288** is omitted from the handle spline **166** when an alternate method of attaching the recliner handle **14** to the handle spline **166** is selected.
(49) Referring to FIGS. **6**A and **6**B, the handle spline **166** includes a cylindrical ring **290** projecting from the splined portion **270**. The cylindrical ring **290** is generally cylindrically-shaped and has a longitudinal axis aligned with the handle spline **166** longitudinal axis **34**B. Further, the cylindrical ring **290** is spaced apart from the upper end surface **278** and the lower end surface **282** of the handle spline **166**. The handle channel **288** is also spaced apart from the cylindrical ring **290** and positioned between the cylindrical ring **290** and the upper end surface **278** of the handle spline **166**. The cylindrical ring **290** has an outer cylindrical surface **294** having an outer diameter **290**A greater than a maximum diameter **270**A of the splined portion **270**. Further, the cylindrical ring **290** includes opposing upper and lower surfaces **298**, **298**′ extending between respective upper and lower edges **294**A, **294**B of the outer cylindrical surface **294** and the outer surface **274** of the splined portion **270**. The lower surface **298**′ of the cylindrical ring **290** is spaced apart from the lower end surface **282** of the handle spline **166** such that the lower surface **298**′ of the cylindrical ring **290** frictionally engages with the upper bracket surface **190** of the spring bracket **158** when the lower end surface **282** of the handle spline **166** is inserted into the alignment aperture **250** of the spring bracket **158**, as shown in FIG. **12**.
(50) Referring to FIGS. **6**A and **12**, the handle spline **166** includes an elongated disc shaft aperture **310** aligned with the longitudinal axis **34**B of the handle spline **166** with a lower opening **314** extending through the lower end surface **282** of the handle spline **166**. The disc shaft aperture **310** is sized and shaped to matingly engage with the disc shaft **62**. Thus, the disc shaft aperture **310** has an elongated inner aperture portion **310**A sized and shaped to matingly engage with the elongated shaft end portion **66** of the disc shaft **62** while maintaining about 6 degrees of loss motion between the disc shaft **62** and the disc shaft aperture **310**. It is understood that the amount of degrees of loss motion between the disc shaft **62** and the disc shaft aperture **310** can be greater or less than about 6 degrees without varying the scope of the invention. Further, the disc shaft aperture **310** has an elongated outer aperture portion **310**B sized and shaped to matingly engage with the central cylindrical portion **98** of the disc shaft **62**. In addition, the disc shaft aperture **310** has an upper end surface **310**C configured to frictionally engage with the distal end surface **80** of the disc shaft **62** when assembled as part of the overload protection device **10**.
(51) Referring to FIGS. **7** through **9**, there is about 6 degrees of lost motion between the handle spline **166** and the disc shaft **62** when the handle spline **166** is rotated. FIG. **7** shows a cross-

sectional view of the handle spline **166** and the disc shaft **62** taken along section line **7-7** of FIG. **1** showing the disc shaft **62** and the handle spline **166** in a neutral position **322**. The neutral position **322** shown in FIG. **7** corresponds to the disc shaft **62** being in the home angular position **36**A with the recliner handle **14** being unrestrained and aligned with the home position **14**A, as illustrated in FIGS. **1** and **3**. As shown in FIG. **7**, the elongated shaft end portion **66** of the disc shaft **62** has opposing shaft flat side portions **72**, **74** and shaft curved side portions **78**, **78′** extending between the opposing shaft flat side portions **72**, **74**.

(52) In contrast, as shown in FIG. **7**, the cross-sectional shape of the inner aperture portion **310**A of the disc shaft aperture **310** in the handle spline **166** has a bowtie-shaped appearance, more generally described as an irregular concave hexagonal shape. The inner aperture portion **310**A includes opposing first and second side walls **324**, **325** and opposing first and second end walls **326**, **327**. The first side wall **324** comprises a first side portion **328** extending at an angle from a second side portion **328′** such that an interior angle **328**A between the first side portion **328** and the second side portion **328′** is greater than 180 degrees, as measured within the disc shaft aperture **310**. Similarly, the second side wall **325** includes a third side portion **329** extending at an angle from a fourth side portion **329′** such that an interior angle **329**A between the third side portion **329** and the fourth side portion **329′** is greater than 180 degrees, as measured within the disc shaft aperture **310**. Further, the first side portion **328** is generally parallel to and spaced apart from the fourth side portion **329′**. Similarly, the second side portion **328′** is generally parallel to and spaced apart from the third side portion **329**. In addition, the first side portion **328** is non-parallel to the second side portion **328′**. Likewise, the third side portion **329** is non-parallel to the fourth side portion **329′**. In certain embodiments, one or more of the first, second, third, and fourth side portions **328**, **328′**, **329**, **329′** include curved surfaces, tapered surfaces, and/or generally flat surfaces.

(53) Also shown in FIG. **7**, extending between ends **328**B, **329**B of the first side portion **328** and the third side portion **329** of the disc shaft aperture **310** is the first end wall **326**. Likewise, extending between ends **328**C, **329**C of the second side portion **328′** and the fourth side portion **329′** is the second end wall **327**. In the embodiment shown in FIG. **7** the first and second end walls **326**, **327** are curved walls. The inner aperture portion **310**A of the disc shaft aperture **310** is sized and shaped such that a first junction **331** of the first side portion **328** and the second side portion **328′** and a second junction **331′** of the third side portion **329** and the fourth side portion **329′** frictionally engage the respective adjacent shaft flat side portions **72**, **74** of the disc shaft **62** when assembled. It is understood that in certain embodiments the first and second junctions **331**, **331′** are spaced apart from the adjacent shaft flat side portions **72**, **74** without altering the scope of the invention. When the disc shaft **62** is centered within the disc shaft aperture **310** in the neutral position **322** shown in FIG. **7**, each of the first through fourth side portions **328**, **328′**, **329**, **329′** of the disc shaft aperture **310** taper away from the adjacent shaft flat side portions **72**, **74** of the disc shaft **62** with an approximate loss motion angle **332**, **332′**. Thus, each of the first through fourth side portions **328**, **328′**, **329**, **329′** of the disc shaft aperture **310** are essentially disengaged from the adjacent shaft flat side portions **72**, **74** of the disc shaft **62** when the handle spline **166** and the disc shaft **62** are in the neutral position **322** shown in FIG. **7**. The approximate loss motion angle **332**, **332′** represents the loss motion between the handle spline **166** and the disc shaft **62** when the handle spline **166** is rotated.

(54) In order to engage the disc shaft aperture **310** of the handle spline **166** with the disc shaft **62**, the handle spline **166** is rotated through approximately the loss motion angle **332**, **332′**, as illustrated in FIGS. **8** and **9**. Once the disc shaft aperture **310** has fully engaged with the disc shaft **62**, additional rotation of the handle spline **166** results in the disc shaft **62** rotating with the handle spline **166**.

(55) When the recliner handle **14** is in the home position **14**A shown in FIG. **1**, the handle spline **166** and the disc shaft **62** are in the neutral position **322** with the second side portion **328′** of the

disc shaft aperture **310** disengaged from the adjacent shaft flat side portion **72** of the disc shaft **62**, as shown in FIG. **7**. More specifically, the second side portion **328′** of the disc shaft aperture **310** tapers away from the adjacent shaft flat side portion **72** of the disc shaft **62** by the loss motion angle **332** when the handle spline **166** and the disc shaft **62** are in the neutral position **322**.

(56) Upward torque applied to the recliner handle **14** rotates the recliner handle **14** and the attached handle spline **166** in the clockwise direction A illustrated in FIG. **1**. The clockwise rotation A of the recliner handle **14** rotates the handle spline **166** in a clockwise direction **333** shown in FIG. **8**. Initial rotation of the handle spline **166** less than the loss motion angle **332** does not result in rotation of the disc shaft **62** because the disc shaft aperture **310** is disengaged from the disc shaft **62**. Thus, applying torque in the clockwise direction A to the recliner handle **14** and rotating the recliner handle **14** less than the loss motion angle **332** results in the handle spline **166** rotating independent of the disc shaft **62** since the disc shaft **62** is disengaged from the disc shaft aperture **310**. The disc shaft **62** engages with the disc shaft aperture **310** when the recliner handle **14** and the attached handle spline **166** rotate approximately the loss motion angle **332** in the clockwise direction **333**. The disc shaft **62** rotates with the handle spline **166** after the recliner handle **14** and the handle spline **166** rotate more than the loss motion angle **332**.

(57) The disc shaft **62** is engaged with the disc shaft aperture **310**, as illustrated in FIG. **8**, when the recliner handle **14** is spaced apart from the home position **14**A by at least the loss motion angle **332**. Referring to FIG. **8**, as the handle spline **166** is rotated in the clockwise direction **333** from the neutral position **322** shown in FIG. **7**, the second side portion **328′** of the disc shaft aperture **310** is pivoted towards the adjacent shaft flat side portion **72** of disc shaft **62**. The second side portion **328′** of the disc shaft aperture **310** frictionally engages with the adjacent shaft flat side portion **72** of the disc shaft **62** when the handle spline **166** is rotated through the loss motion angle **332** in the clockwise direction **333**. The third side portion **329** of the disc shaft aperture **310** likewise frictionally engages with the adjacent shaft flat side portion **74** when the handle spline **166** is rotated in the clockwise direction **333** since the third side portion **329** and the second side portion **328′** of the disc shaft aperture **310** are generally parallel. Additional rotation of the handle spline **166** in the clockwise direction **333** causes the disc shaft **62** to rotate with the handle spline **166** since the second and third side portions **328′**, **329** of the disc shaft aperture **310** have engaged with the adjacent shaft flat side portions **72**, **74** of the disc shaft **62**. Thus, the loss motion angle **332** illustrates the amount of loss motion between the handle spline **166** and the disc shaft **62** when the handle spline **166** is rotated in the clockwise direction **333** from the neutral position **322**.

(58) As shown in FIG. **7**, the fourth side portion **329′** of the disc shaft aperture **310** tapers away from the adjacent shaft flat side portion **74** of the disc shaft **62** by the loss motion angle **332′** when the handle spline **166** and the disc shaft **62** are in the neutral position **322**. When the handle spline **166** is rotated in a counterclockwise direction **333′** from the neutral position **322** of FIG. **7**, the fourth side portion **329′** is pivoted towards the adjacent shaft flat side portion **74** of disc shaft **62**, as illustrated in FIG. **9**. The fourth side portion **329′** frictionally engages with the adjacent shaft flat side portion **74** of the disc shaft **62** when the handle spline **166** is rotated through the loss motion angle **332′** in the counterclockwise direction **333′**. The first side portion **328** of the disc shaft aperture **310** likewise frictionally engages with the adjacent shaft flat side portion **72** when the handle spline **166** is rotated in the counterclockwise direction **333′** since the fourth side portion **329′** and the first side portion **328** of the disc shaft aperture **310** are generally parallel. Additional rotation of the handle spline **166** in the counterclockwise direction **333′** causes the disc shaft **62** to rotate with the handle spline **166** since the first and fourth side portions **328**, **329′** of the disc shaft aperture **310** have engaged with the adjacent shaft flat side portions **72**, **74** of the disc shaft **62**. Thus, the loss motion angle **332′** illustrates the amount of loss motion between the handle spline **166** and the disc shaft **62** when the handle spline **166** is rotated in the counterclockwise direction **333′** from the neutral position **322**. In certain embodiments, the loss motion angle **332** is approximately equal to the loss motion angle **332′**. However, it will be appreciated that the loss

motion angle **332** can be greater than or less than loss motion angle **332′** without varying the scope of the invention. Further, the clockwise direction **333** and the counterclockwise direction **333′** of the handle spline **166**, as viewed in FIGS. **8** and **9**, is generically described as first and second rotational directions **333**, **333′**, respectively.

(59) The handle spline **166** has a clinch ring slot **334** extending radially through part of the cylindrical ring **290**, as shown in FIGS. **6**A and **12**. The clinch ring slot **334** is configured to allow insertion of the clinch ring **162** into the clinch ring slot **334**. Further, the clinch ring slot **334** is sized and shaped such that the clinch ring **162** is aligned with the channel **106** in the disc shaft **62** when the spring bracket **158** and the handle spline **166** are assembled on the disc shaft **62**, as shown in FIG. **12**. The clinch ring **162**, the clinch ring slot **334**, and the channel **106** in the disc shaft **62** are sized and shaped such that insertion of the clinch ring **162** into the clinch ring slot **334** does not restrict the rotational movement of the handle spline **166** with respect to the disc shaft **62**. In particular, the clinch ring **162** does not restrict the rotation of the handle spline **166** with respect to the disc shaft **62** while the handle spline **166** is rotated in the first and second rotational directions **333**, **333′** through at least an angle equal or greater than the loss motion angle **332**, **332′** from the neutral position **322** shown in FIG. **7**. It will be appreciated that the size and shape of the clinch ring slot **334**, the channel **106** in the disc shaft **62**, and the clinch ring **162** may vary without altering the scope of the invention.

(60) As shown in FIG. **6**A, the clinch ring slot **334** includes opposing upper and lower surfaces **334**A, **334**B as well as opposing side surfaces **334**C, **334**D. The disc shaft aperture **310** forms an upper opening **338** in the upper surface **334**A of the clinch ring slot **334**. A lower opening **338′** is formed by the disc shaft aperture **310** in the lower surface **334**B of the clinch ring slot **334**.

(61) Referring to FIG. **2**, the handle inertia spring **170** is a coil spring having a coiled portion **346** having a central passageway **348**, a first spring end **350** extending from an upper side **346**A of the handle inertia spring **170**, and a second spring end **350′** extending from a lower side **346**B of the handle inertial spring **170**. The handle inertia spring **170** is formed of a metal.

(62) In the embodiment shown in FIG. **2**, the first spring end **350** of the handle inertia spring **170** is bent into an "L" shape having a first retention portion **354** extending at an angle from the handle inertia spring **170**. The first spring end **350** and the first retention portion **354** are sized and shaped such that the first spring end **350** can be passed through the spring slot **214** in the spring bracket **158**. The first retention portion **354** retains the first spring end **350** in frictional engagement with the spring bracket **158** when the handle inertia spring **170** is assembled as part of the overload protection device **10**, as illustrated in FIG. **14**.

(63) Also shown in FIG. **2**, the second spring end **350′** of the handle inertia spring **170** includes a bent portion **358** generally extending in the direction of the axis of rotation **34** of the recliner **18** after assembly with the spring bracket **158** and the recliner B-bracket **44**. A second retention portion **358′** extends at an angle from the bent portion **358** of the handle inertia spring **170**. As shown in FIG. **14**, the second retention portion **358′** and the bent portion **358** of the handle inertia spring **170** are sized and shaped such that the second spring end **350′** can be passed through a spring retention hole **366** in the recliner B-bracket **44** when the handle inertia spring **170** is assembled as part of the overload protection device **10**. More specifically, the second spring end **350′** is configured to pass through the spring retention hole **366** in the recliner B-bracket **44** when the handle spline **166** and the second flange **207** of the spring bracket **158** are passed through the central passageway **348** of the coiled portion **346**, as illustrated in FIG. **13**. Further, the second retention portion **358′** and the bent portion **358** are sized and shaped such that the second retention portion **358′** retains the second spring end **350′** through the spring retention hole **366** after the second spring end **350′** is inserted through the spring retention hole **366** in the recliner B-bracket **44**. It will be appreciated that the size and shape of the handle inertia spring **170**, including the first and second spring ends **350**, **350′**, may vary without altering the scope of the invention.

(64) The tab **174** formed on the recliner B-bracket **44** is shown in FIGS. **2** and **14**. Referring to FIG.

2, the recliner B-bracket **44** includes a mounting portion **44**A configured to matingly engage with bosses **46**, **48** projecting from the guide plate **42**A of the disc recliner assembly **42**. In various embodiments, the mounting portion **44**A is integrally formed with the recliner B-bracket **44**. In alternate embodiments, the mounting portion **44**A is a separate component assembled with the recliner B-bracket **44**. Referring to FIG. **14**, the recliner B-bracket **44** and the mounting portion **44**A can be formed out of a metal, a plastic, and/or combinations thereof. Preferably, at least the mounting portion **44**A of the recliner B-bracket **44** is formed from a metal. In certain embodiments, the mounting portion **44**A is a metal bracket that is fixedly coupled to the recliner B-bracket **44**.
(65) Referring to FIGS. **2** and **14**, the tab **174** formed on the recliner B-bracket **44** is bent away from a disc-shaped portion **44**B of the mounting portion **44**A of the recliner B-bracket **44**. In alternate embodiments, the tab **174** is a separate component fixedly coupled to the mounting portion **44**A and/or to the recliner B-bracket **44**. The tab **174** includes an overload stop **174**A formed by an edge surface of the tab **174**. The overload stop **174**A is offset from the disc-shaped portion **44**B of the mounting portion **44**A. The tab **174** is sized and shaped such that the stop surface **246** of the stop flange **208** of the spring bracket **158** will frictionally engage the overload stop **174**A of the tab **174** when the spring bracket **158** and handle spline **166** are assembled on the disc shaft **62** and the spring bracket **158** rotated towards the tab **174** in the second rotational direction B shown in FIG. **14**. It will be appreciated that the size and shape of the recliner B-bracket **44**, the mounting portion **44**A, the tab **174**, and the overload stop **174**A may vary without altering the scope of the invention.
(66) FIGS. **10** through **13** illustrate an installation process of the overload protection device **10** onto the recliner **18**. Referring to FIG. **10**, the recliner **18** is shown assembled with the recliner B-bracket **44** and the seat back **26**. The recliner **18** includes the disc recliner assembly **42** comprising the guide plate **42**A, the tooth plate **42**B, and the disc shaft **62** as well as other components. Further, the return spring **38** is shown assembled with the disc recliner assembly **42** in FIG. **10**. The lower end surface **282** of the handle spline **166** is inserted into the alignment aperture **250** in the spring bracket **158** with the plurality of splines **286** of the handle spline **166** aligned with the plurality of splines **254** in the alignment aperture **250**, as illustrated by arrow D shown in FIG. **10**. The handle spline **166** is assembled with the disc shaft **62** after the handle spline **166** is assembled with the spring bracket **158**, as illustrated by arrow E shown in FIG. **10**. Alternatively, the spring bracket **158** can be assembled with the disc shaft **62** prior to the handle spline **166** being inserted into the alignment aperture **250** of the spring bracket **158**. In both cases, a portion of the disc shaft **62** passes through the alignment aperture **250** of the spring bracket **158**.
(67) FIG. **11** shows the handle spline **166** and the spring bracket **158** assembled on the disc shaft **62** of the disc recliner assembly **42**. The clinch ring **162** is inserted into the clinch ring slot **334** in the handle spline **166**, as illustrated by arrow F shown in FIG. **11**. The clinch ring **162** axially retains the handle spline **166** on the disc shaft **62**.
(68) A cross-sectional view of the assembly of the disc shaft **62**, the spring bracket **158**, the handle spline **166**, and the clinch ring **162** is shown in FIG. **12**. Referring to FIG. **12**, the lower bracket surface **194** of the spring bracket **158** abuts the base end face **94** of the disc shaft **62**. In addition, the lower end surface **282** of the handle spline **166** either abuts the base end face **94** of the disc shaft **62** or the lower end surface **282** is spaced apart from the base end face **94** of the disc shaft **62**, depending on specific dimensions of the individual components. Further, the lower end surface **282** of the handle spline **166** is positioned within the alignment aperture **250** in the spring bracket **158**. The spring bracket **158** will rotate with the handle spline **166** since the plurality of splines **254** in the alignment aperture **250** are matingly engaged with the plurality of splines **286** of the handle spline **166**.
(69) While the spring bracket **158** and the handle spline **166** are shown assembled with the disc shaft **62** in FIG. **12**, the handle spline **166** is not fixedly coupled to the disc shaft **62**. The handle spline **166** is selectively coupled to the disc shaft **62** based in part on the rotational position of the

handle spline **166** with respect to the disc shaft **62**. As shown in FIGS. **7** through **9**, there is about 6 degrees of lost motion between the handle spline **166** and the disc shaft **62** (illustrated by the loss motion angles **332**, **332**′) when the handle spline **166** is rotated from the neutral position **322**. It is understood that the magnitude of the loss motion angles **332**, **332**′ can vary without altering the scope of the invention.

(70) Also shown in FIG. **12**, the lower surface **298**′ of the cylindrical ring **290** of the handle spline **166** abuts the upper bracket surface **190** of the spring bracket **158**. The clinch ring slot **334** in the handle spline **166** is aligned with the channel **106** of the disc shaft **62** such that the clinch ring **162** can be inserted into the clinch ring slot **334**. Preferably, a portion of the inner cylindrical surface **262** of the clinch ring **162** is positioned within the channel **106**, preventing movement of the handle spline **166** in the direction of the axis of rotation **34**.

(71) After the spring bracket **158** and the handle spline **166** are assembled with the disc shaft **62**, the handle inertia spring **170** is attached to the spring bracket **158** and the recliner B-bracket **44** as illustrated in FIG. **13**. Referring to FIG. **13**, a first portion **346**′ of the coiled portion **346** of the handle inertia spring **170** is positioned underneath the distal flange end portion **230** of the second flange **207** of the spring bracket **158** such that the upper surface **346**A of the coiled portion **346** abuts the distal flange end portion **230** of the spring bracket **158**. A second portion **346**″ of the coiled portion **346** of the handle inertia spring **170** is placed on the spring bracket **158** such that the lower side **346**B of the coiled portion **346** abuts the upper bracket surface **190** of the main bracket portion **186** of the spring bracket **158** with the handle spline **166** positioned in the central passageway **348** of the handle inertia spring **170**. The first spring end **350** of the handle inertia spring **170** is slid through the spring slot **214** in the first flange **206** of the spring bracket **158**. The second spring end **350**′ of the handle inertia spring **170** is passed through the spring retention hole **366** in the recliner B-bracket **44**. The spring torque of the handle inertia spring **170** retains the first retention portion **354** of the first spring end **350** of the handle inertia spring **170** in frictional engagement with the first flange **206** of the spring bracket **158**. In addition, the spring torque of the handle inertia spring **170** retains the second retention portion **358**′ of the second spring end **350**′ of the handle inertia spring **170** in frictional engagement with the recliner B-bracket **44**.

(72) Referring to FIG. **14**, the spring torque within the handle inertia spring **170** rotationally biases the spring bracket **158** and handle spline **166** towards the home angular position **36**A in the second rotational direction B. The stop surface **246** on the spring bracket **158** is frictionally engaged with the overload stop **174**A on the recliner B-bracket **44** when the handle spline **166** is in the home angular position **36**A. More specifically, the return spring **38** which is operatively coupled between the recliner **18** and the disc shaft **62** biases the disc shaft **62** towards the home angular position **36**A. As such, the return spring **38** automatically rotates the disc shaft **62** towards the home angular position **36**A when the disc shaft **62** is unloaded by an externally-applied load. Further, the handle inertia spring **170** rotationally biases the spring bracket **158** towards the home angular position **36**A. When the stop surface **246** on the spring bracket **158** frictionally engages with the overload stop **174**A on the recliner B-bracket **44**, and the disc shaft **62** is in the home angular position **36**A, the disc shaft aperture **310** of the handle spline **166** is automatically positioned in the neutral position **322** with the disc shaft **62** disengaged from the disc shaft aperture **310**.

(73) FIG. **15** shows a cutaway view of the disc shaft **62** and the handle spline **166** of FIG. **14** illustrating the spring bracket **158** and the handle spline **166** in the home angular position **36**A with the disc shaft **62** disengaged from the disc shaft aperture **310** of the handle spline **166**. The home angular position **36**A shown in FIG. **15** corresponds to the neutral position **322** of the disc shaft **62** and the handle spline **166** shown in FIG. **7**. Referring to FIG. **15**, when the handle spline **166** is unloaded (i.e., an externally-applied load is not applied to the recliner handle **14** attached to the handle spline **166**), the spring torque in the handle inertia spring **170** ensures that the spring bracket **158** and the coupled handle spline **166** are in the home angular position **36**A. If the handle spline **166** is spaced apart from the home angular position **36**A and the recliner handle **14** is unrestrained,

the handle inertial spring **170** automatically rotates the spring bracket **158** in the second rotational direction B until the stop surface **246** of the stop flange **208** frictionally engages the overload stop **174**A of the recliner B-bracket **44**.

(74) Referring to FIGS. **14** and **15**, the return spring **38** of the recliner **18** rotationally biases the disc shaft **62** towards the home angular position **36**A in the second rotational direction B when the disc shaft **62** is unloaded. The return spring **38** ensures that the recliner **18** is placed in the locked condition when upward torque is not applied to the recliner handle **14**. When the recliner **18** includes a cam that locks and unlocks the recliner **18**, rotating the disc shaft **62** in the first rotational direction A is alternatively described as a cam unlocking direction A. Similarly, rotating the disc shaft **62** in the second rotational direction B is alternatively described as a cam locking direction B.

(75) During normal use, upward torque (first rotational direction A) is applied to the recliner handle **14** to initiate normal recline operation of the vehicle seat **22**, as shown in FIG. **1**. Rotation of the recliner handle **14** in the first rotational direction A applies torque to the handle spline **166** causing the handle spline **166** to be rotated in a clockwise direction (arrow G), as viewed in FIGS. **14** and **15**. As illustrated in FIGS. **7** and **8**, there is lost motion present between the handle spline **166** and the disc shaft **62** of about 6 degrees. It will be understood that the loss motion between the handle spline **166** and the disc shaft **62** can be greater or less than about 6 degrees without varying the scope of the invention. Referring to FIGS. **7**, **8**, and **15**, the handle spline **166** rotates in the clockwise direction (arrow **333** in FIG. **8**, arrow Gin FIG. **15**) through approximately the loss motion angle **332** before the second and third side portions **328′**, **329** of the disc shaft aperture **310** frictionally engage the respective shaft flat side portions **72**, **74** of the disc shaft **62**. This initial rotation of the handle spline **166** takes up the looseness between the disc shaft aperture **310** and the disc shaft **62**. The disc recliner assembly **42** functions as a typical direct drive disc recliner assembly after looseness between the disc shaft aperture **310** and the disc shaft **62** is taken up by the initial rotation of the handle spline **166**.

(76) Additional rotation of the handle spline **166** in the clockwise direction (arrow **333** in FIG. **8**, arrow Gin FIG. **15**) past the loss motion angle **332** causes the disc shaft **62** to rotate with the handle spline **166**. Rotation of the recliner handle **14** in the first rotational direction A shown in FIG. **1** results in the handle spline **166**, the spring bracket **158**, and the disc shaft **62** being rotated to the release angular position **36**B shown in FIG. **16**. When the recliner handle **14** is rotated in the first rotational direction A, rotational torque (arrow G in FIG. **16**) is applied to the handle spline **166**, retaining the handle spline **166** in the release angular position **36**B. In certain embodiments, the handle spline **166** is rotated past the release angular position **36**B when the recliner handle **14** is rotated further in the first rotational direction A.

(77) When the handle spline **166** is in the release angular position **36**B shown in FIG. **16**, releasing the recliner handle **14** results in the handle inertia spring **170** automatically rotating the spring bracket **158** in the counterclockwise direction (arrow H) towards the home angular position **36**A, as shown in FIG. **17**. The handle spline **166** rotates with the spring bracket **158** since the plurality of splines **286** of the handle spline **166** are matingly engaged with the plurality of splines **254** in the alignment aperture **250** of the spring bracket **158**. Spring torque in the handle inertia spring **170** retains the stop surface **246** on the spring bracket **158** in frictional engagement with the overload stop **174**A of the recliner B-bracket **44**.

(78) FIGS. **18** through **20** illustrate operation of the overload protection device **10** during abnormal use. Referring to FIG. **1**, abnormal use occurs when a load is applied to the recliner handle **14** in the abuse rotation direction C when the recliner handle **14** is in the home position **14**A. The handle spline **166** and the disc shaft **62** are shown in the home angular position **36**A in FIG. **18** with the disc shaft **62** disengaged from the disc shaft aperture **310**. The relative rotational positions of the handle spline **166** and the disc shaft **62** shown in FIG. **18** is alternatively described as the neutral position **322**.

(79) The relative rotational positions of the handle spline **166** and the disc shaft **62** in the neutral position **322** shown in FIG. **18** correspond to the recliner handle **14** being in the home position **14**A shown in FIG. **1**. The relative rotational position of the disc shaft aperture **310** with respect to the disc shaft **62** of FIG. **18** also corresponds to the relative rotational position of the disc shaft aperture **310** with respect to the disc shaft **62** shown in FIG. **7**. In specific, when the handle spline **166** and the disc shaft **62** are in the neutral position **322** shown in FIGS. **7** and **18**, the fourth side portion **329**′ of the disc shaft aperture **310** in the handle spline **166** tapers away from the adjacent shaft flat side portion **74** of the disc shaft **62**. As such, the disc shaft aperture **310** is disengaged from the disc shaft **62** when the handle spline **166** is in neutral position **322** with no load applied to the recliner handle **14**.
(80) Referring to FIGS. **1** and **19**, when the recliner handle **14** is in the home position **14**A shown in FIG. **1** and a load is applied to the recliner handle **14** in the abuse rotation direction C, abuse torque J (FIG. **19**) is applied to the handle spline **166**. The abuse torque J applied to the handle spline **166** in the cam locking direction B is represented by arrow J shown in FIG. **19**. An exemplary amount of abuse torque J is approximately 120 Nm applied to the recliner handle **14** in a cam locking direction B when the recliner handle **14** is in the home position **14**A.
(81) Referring to FIG. **19**, the abuse torque J applied to the handle spline **166** urges the handle spline **166** to rotate past the home angular position **36**A in the abuse rotation direction C. For typical disc recliners lacking the overload protection device **10**, the abuse torque J applied to the handle spline **166** causes the handle spline **166** to rotate in the abuse rotation direction C past the home angular position **36**A and causes the typical disc recliner to be overlocked. However, as shown in FIG. **19**, the stop surface **246** on the spring bracket **158** of the overload protection device **10** is frictionally engaged with the overload stop **174**A on the recliner B-bracket **44** when the handle spline **166** is in the neutral position **322**. Further, the handle spline **166** is matingly engaged with the spring bracket **158** and rotationally travels with the spring bracket **158**. More specifically, the plurality of splines **286** of the handle spline **166** are matingly engaged with the plurality of splines **254** in the alignment aperture **250** of the spring bracket **158** such that the handle spline **166** rotates with the spring bracket **158**, as illustrated in FIGS. **10** and **12**. Thus, abuse torque J applied to the handle spline **166** initiates rotation of the handle spline **166** in the abuse rotation direction C, causing the stop surface **246** on the stop flange **208** of the spring bracket **158** to impact the overload stop **174**A on the tab **174** of the recliner B-bracket **44**, as shown in FIG. **19**. The engagement between the stop flange **208** and the overload stop **174**A prevents rotation of the handle spline **166** in the abuse rotation direction C past the home angular position **36**A. Referring to FIG. **20**, the abuse torque J applied to the handle spline **166** is transferred to the spring bracket **158** and applies load K (illustrated by arrow K) to the overload stop **174**A on the tab **174** of the recliner B-bracket **44**. This effectively transfers the abuse torque J applied to the handle spline **166** to the tab **174** of the recliner B-bracket **44**.
(82) As shown in FIGS. **7**, **9** and **20**, the looseness between the disc shaft **62** and the disc shaft aperture **310** in the handle spline **166** prevents the abuse torque J from being transferred to the disc shaft **62**. FIG. **7** shows the handle spline **166** and the disc shaft **62** in the neutral position **322** with the first side portion **328** and the fourth side portion **329**′ of the disc shaft aperture **310** tapering away from the adjacent shaft flat side portions **72**, **74**. FIG. **9** shows the handle spline **166** rotated in the counterclockwise direction **333**′ from the neutral position **322** shown in FIG. **7** through approximately the loss motion angle **332**′ with the overload stop **174**A omitted from the recliner B-bracket **44**. The abuse rotation direction C in FIG. **20** corresponds to counterclockwise direction **333**′ shown in FIG. **9**. Referring to FIG. **9**, the handle spline **166** would have to rotate in the counterclockwise direction **333**′ (abuse rotation direction C of FIG. **20**) through approximately the loss motion angle **332**′ prior to the fourth side portion **329**′ of the disc shaft aperture **310** frictionally engaging the adjacent shaft flat side portion **74** of the disc shaft **62**. The disc shaft **62** rotates with the handle spline **166** when the handle spline **166** rotates past the loss motion angle

332′ in the counterclockwise direction **333**′ since the first and fourth side portions **328**, **329**′ have frictionally engaged the respective adjacent shaft flat side portions **72**, **74**.

(83) However, as shown in FIG. **20**, frictional engagement between the stop surface **246** on the stop flange **208** of the spring bracket **158** and the overload stop **174**A on the tab **174** of the recliner B-bracket **44** prevents rotation of the handle spline **166** in the abuse rotation direction C past the home angular position **36**A. The abuse torque J applied to the recliner handle **14** is transferred through the handle spline **166** and loads up the spring bracket **158** against the tab **174** of the recliner B-bracket **44**. The looseness between the disc shaft aperture **310** in the handle spline **166** and the disc shaft **62** prevents transferring the abuse torque J from the handle spline **166** to the disc shaft **62**. Since the abuse torque J is transferred to the recliner B-bracket **44** instead of being transferred to the disc shaft **62**, the disc shaft **62** is not rotated past the home angular position **36**A when downward torque is applied to the recliner handle **14**. The overload protection device **10** prevents the disc shaft **62** from being rotated in the abuse rotation direction C past the home angular position **36**A, effectively preventing the recliner **18** from being overlocked. Further, internal components of the disc recliner assembly **42** are not directly loaded by downward torque of the recliner handle **14** since load applied by the abuse torque J is transferred between the recliner handle **14**, the handle spline **166**, the spring bracket **158**, and the tab **174** of the recliner B-bracket **44**.

(84) A second embodiment of a recliner handle **14** with an overload protection device **10**′ is illustrated in FIGS. **21** through **26**. The first embodiment of the overload protection device **10** illustrated in FIG. **2** includes a spring bracket **158** having a stop flange **208** that is assembled with a separate handle spline **166**. In contrast, the overload protection device **10**′ shown in FIGS. **21** through **26** has a stop flange **208**′ integrated with the handle spline **166**′. In addition, the handle inertia spring **170** shown in FIG. **2** is replaced with a handle inertia spring **170**′ that is operatively connected between the handle spline **166**′ and the recliner B-bracket **44**, as shown in FIG. **21**.

(85) FIG. **22** shows an exploded view of components of the second embodiment of the overload protection device **10**′. The handle spline **166**′ includes a disc shaft aperture **310**′ passing longitudinally through the handle spline **166**′. The handle spline **166**′ is assembled with the recliner **18** by inserting the disc shaft **62** of the recliner **18** into the disc shaft aperture **310**′. The handle inertia spring **170**′ of the second embodiment includes a first spring end **350**A configured to frictionally engage with a first spring retention surface **370** on handle spline **166**′. Further, a second spring end **350**B of the handle inertia spring **170**′ is configured to frictionally engage with a spring retention flange **366**B on the recliner B-bracket **44**, as shown in FIG. **21**.

(86) FIG. **23** shows the handle spline **166**′ assembled with the recliner B-bracket **44** and the disc shaft **62** of the second embodiment, illustrating the integrated stop flange **208**′ of the handle spline **166**′ frictionally engaged with the overload stop **174**A of the recliner B-bracket **44**. A comparison of the first embodiment shown in FIG. **15** and the second embodiment shown in FIG. **23** illustrates similarities between the first and second embodiments. For example, the stop flange **208** of the spring bracket **158** rotates with the handle spline **166** when the handle spline **166** is rotated, as shown and described with respect to FIGS. **15** through **17**. Referring to FIG. **23**, the stop flange **208**′ of the second embodiment rotates with the handle spline **166**′ since the stop flange **208**′ is integrally formed with the handle spline **166**′.

(87) In addition, the disc shaft aperture **310**′ of the handle spline **166**′ of the second embodiment is shaped and sized similarly to the disc shaft aperture **310** of the handle spline **166** in the first embodiment, as shown by comparing FIG. **24** and FIG. **7**. Referring to FIG. **24**, the disc shaft aperture **310**′ of the handle spline **166**′ of the second embodiment has an inner aperture portion **310**A′ sized and shaped to matingly engage with the elongated shaft end portion **66** of the disc shaft **62** while maintaining about 6 degrees of loss motion between the disc shaft **62** and the disc shaft aperture **310**′. It is understood that the amount of degrees of loss motion between the disc shaft **62** and the disc shaft aperture **310**′ can be greater or less than about 6 degrees without varying the

scope of the invention. The inner aperture portion **310**A′ of the disc shaft aperture **310**′ includes opposing first and second side walls **324**′, **325**′ and opposing first and second end walls **326**′, **327**′. The first side wall **324**′ comprises a first side portion **328**D extending at an angle from a second side portion **328**D′ such that an interior angle **328**A′ between the first side portion **328**D and the second side portion **328**D′ is greater than 180 degrees, as measured within the disc shaft aperture **310**′. Similarly, the second side wall **325**′ includes a third side portion **329**D extending at an angle from a fourth side portion **329**D′ such that an interior angle **329**A′ between the third side portion **329**D and the fourth side portion **329**D′ is greater than 180 degrees, as measured within the disc shaft aperture **310**′. Further, the first side portion **328**D is generally parallel to and spaced apart from the fourth side portion **329**D′. Similarly, the second side portion **328**D′ is generally parallel to and spaced apart from the third side portion **329**D. In addition, the first side portion **328**D is non-parallel to the second side portion **328**D′. Likewise, the third side portion **329**D is non-parallel to the fourth side portion **329**D′. In certain embodiments, one or more of the first, second, third, and fourth side portions **328**D, **328**D′, **329**D, **329**D′ include curved surfaces, tapered surfaces, and/or generally flat surfaces. Also shown in FIG. **24**, extending between ends **328**B′, **329**B′ of the first side portion **328**D and the third side portion **329**D of the disc shaft aperture **310**′ is the first end wall **326**′. Likewise, extending between ends **328**C′, **329**C′ of the second side portion **328**D′ and the fourth side portion **329**D′ is the second end wall **327**′. In the embodiment shown in FIG. **24** the first and second end walls **326**′, **327**′ are curved walls.

(88) FIG. **26** shows a cutaway perspective view of the handle spline **166**′ of the second embodiment assembled with the disc shaft **62**. In FIG. **26**, the disc shaft **62** and the disc shaft aperture **310**′ in the handle spline **166**′ are positioned in a neutral position **322** with the disc shaft aperture **310**′ disengaged from the disc shaft **62**. Further, the inner aperture portion **310**A′ of the disc shaft aperture **310**′ is sized and shaped such that a first junction **331**D of the first side portion **328**D and the second side portion **328**D′ and a second junction **331**D′ of the third side portion **329**D and the fourth side portion **329**D′ frictionally engage the respective adjacent shaft flat side portions **72**, **74** of the disc shaft **62** when assembled, as shown in FIG. **26**. It is understood that in certain embodiments the first and second junctions **331**D, **331**D′ are spaced apart from the adjacent shaft flat side portions **72**, **74** of the disc shaft **62** without varying the scope of the invention. When the disc shaft **62** is centered within the disc shaft aperture **310**′ in the neutral position **322** shown in FIG. **26**, each of the first through fourth side portions **328**D, **328**D′, **329**D, **329**D′ of the disc shaft aperture **310**′ taper away from the adjacent shaft flat side portions **72**, **74** of the disc shaft **62** with an approximate loss motion angle **332**, **332**′. Thus, each of the first through fourth side portions **328**D, **328**D′, **329**D, **329**D′ of the disc shaft aperture **310**′ are essentially disengaged from the adjacent shaft flat side portions **72**, **74** of the disc shaft **62** when the handle spline **166**′ and disc shaft **62** are in the neutral position **322** shown in FIG. **26**. The approximate loss motion angle **332**, **332**′ represents the loss motion between the handle spline **166**′ and the disc shaft **62** when the handle spline **166**′ is rotated. As with the first embodiment, rotating the handle spline **166**′ from the neutral position **322** less than the loss motion angle **332** results in the handle spline **166**′ rotating independently of the disc shaft **62**. The disc shaft aperture **310**′ engages with the disc shaft **62** when the handle spline **166**′ rotates approximately the loss motion angle **332** from the neutral position **322** shown in FIG. **26**. The disc shaft **62** rotates with the handle spline **166**′ as long as the disc shaft aperture **310**′ is engaged with the disc shaft **62**.

(89) Referring to FIG. **25**, the handle spline **166**′ has a plurality of splines **286**′ spaced around an outer perimeter **274**A of a splined portion **270**′ configured to matingly engage with the recliner handle **14**. Further, the disc shaft aperture **310**′ extends axially through the handle spline **166**′. Also shown in FIG. **25**, the handle spline **166**′ includes a spring retention flange **370**A having a first spring retention surface **370** configured to frictionally engage and retain the first spring end **350**A of the handle inertia spring **170**′. As shown in FIG. **24**, the spring retention flange **370**A extends circumferentially partially around the handle spline **166**′. In the embodiment shown in FIG. **24**, the

spring retention flange **370**A extends between the stop flange **208′** and the first spring retention surface **370**. It is understood that the stop flange **208′** and the first spring retention surface **370** can vary in size, shape, position, and location, including being separated on the handle spline **166′** without varying the scope of the invention.
(90) The handle spline **166′** shown in FIGS. **24** and **25** is sized and shaped to be formed out of a plastic material using a molding process. An exemplary suitable plastic material is Nylon 6/6 with about 30% glass fill. Alternatively, the handle spline **166′** is formable out of a metal such as a zinc aluminum alloy and similar metals. An exemplary, suitable zinc aluminum alloy is Zamak 5 (aluminum 4%, copper 1%, and zinc 95%). It is understood that the handle spline **166′** can be formed of alternate metals and molded from alternate plastic materials without varying the scope of the invention.
(91) One benefit of the recliner handle **14** with the overload protection device **10**, **10′** is damage to the recliner **18** is prevented when downward torque (abuse torque) is applied to the recliner handle **14** when the recliner handle **14** is in the home position **14**A. A second benefit is the abuse torque applied to the recliner handle **14** is transferred away from the recliner **18**. A third benefit of the overload protection device **10**, **10′** between the recliner handle **14** and the recliner **18** is overloading the recliner **18** is avoided when downward torque is applied to the recliner handle **14**.
(92) The invention has been described in an illustrative manner, and it is to be understood that the terminology, which has been used, is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present invention are possible in light of the above teachings. It is, therefore, to be understood that within the scope of the appended claims, the invention may be practiced other than as specifically described.

Claims

1. An overload protection device for a reclining vehicle seat, the overload protection device comprising: an overload stop fixedly coupled to the vehicle seat; a recliner having a disc shaft, wherein the disc shaft is rotatable to actuate the recliner between a locked condition and an unlocked condition for providing reclining adjustment of the vehicle seat; a handle spline rotatable in a first rotational direction from a first position associated with the locked condition to a second position associated with the unlocked condition, wherein the handle spline defines a disc shaft aperture with the disc shaft received in the disc shaft aperture, wherein the disc shaft and the disc shaft aperture are rotationally decoupled when the handle spline is in the first position and rotationally coupled when the handle spline is in the second position; and a stop flange rotationally coupled to the handle spline and arranged to abut the overload stop when the handle spline is in the first position; wherein engagement between the stop flange and the overload stop prevents rotation of the handle spline in a second rotational direction opposite the first rotational direction and away from the second position when the handle spline is in the first position to prevent an overload force being transferred to the recliner.
2. The overload protection device of claim 1, wherein: said first position and said second position being spaced apart by a lost motion angle.
3. The overload protection device of claim 2, wherein: a handle inertia spring is operatively coupled between said handle spline and the vehicle seat, said handle inertia spring biasing said stop flange towards abutment with said overload stop.
4. The overload protection device of claim 3, wherein: said recliner is in said locked condition when said disc shaft is in a home angular position; and said recliner is in said unlocked condition when said disc shaft is in a release angular position.
5. The overload protection device of claim 4, wherein: said disc shaft is rotatable between said home angular position and said release angular position; rotating said disc shaft in said first rotational direction from said home angular position to said release angular position reconfigures

said recliner from said locked condition to said unlocked condition; and rotating said disc shaft in said second rotational direction from said release angular position to said home angular position reconfigures said recliner from said unlocked condition to said locked condition.

6. The overload protection device of claim 5, further comprising: a recliner handle coupled to the handle spline and rotatable between a home position and a recline release position; wherein when said recliner handle is in said home position, said handle spline is in said first position, and said disc shaft aperture is rotationally decoupled from said disc shaft, applying torque in said first rotational direction to said recliner handle and rotating said recliner handle in said first rotational direction less than said lost motion angle results in said handle spline rotating in said first rotational direction with said disc shaft rotationally decoupled from said disc shaft aperture.

7. The overload protection device of claim 6, wherein: when said recliner handle is spaced apart from said home position by at least said lost motion angle, applying torque in said first rotational direction to said recliner handle and rotating said recliner handle in said first rotational direction results in said handle spline rotating in said first rotational direction with said disc shaft rotationally coupled with said disc shaft aperture; and said disc shaft rotating with said handle spline when said disc shaft is rotationally coupled with said disc shaft aperture.

8. The overload protection device of claim 7, wherein: when said handle spline is in said first position with said stop flange abutting said overload stop, rotating said handle spline in said first rotational direction disengages said stop flange from said overload stop.

9. The overload protection device of claim 8, wherein: when said recliner handle is rotated in said first rotational direction to said recline release position, said handle spline is rotated to said release angular position, said disc shaft aperture rotationally coupled with said disc shaft rotates said disc shaft to said release angular position, and said disc shaft rotated to said release angular position reconfigures said recliner into said unlocked condition.

10. The overload protection device of claim 1, wherein: when said handle spline is in said first position with said disc shaft aperture rotationally decoupled from said disc shaft and when said stop flange is abutting said overload stop, applying torque in said second rotational direction to said handle spline results in load being applied to said overload stop by said stop flange while retaining said disc shaft aperture rotationally decoupled from said disc shaft.

11. The overload protection device of claim 4, wherein said disc shaft aperture has an elongated aperture portion comprising: a first side wall comprising a first portion and a second portion joining at a first junction, said first and second portions having a first end and a second end, respectively, spaced apart from said first junction, and said first and second portions being non-parallel; a second side wall comprising a third portion and a fourth portion joining at a second junction, said third and fourth portions having a third end and a fourth end, respectively, spaced apart from said second junction, and said third and fourth portions being non-parallel; a first end wall adjoining said first end of said first side wall and said third end of said second side wall; and a second end wall adjoining said second end of said first side wall and said fourth end of said second side wall.

12. The overload protection device of claim 11, wherein said elongated aperture portion of said disc shaft aperture comprises: said first portion of said first side wall opposing said third portion of said second side wall, said second portion of said first side wall opposing said fourth portion of said second side wall, said first end of said first portion of said first side wall and said third end of said third portion of said second side wall being spaced further apart than said first junction and said second junction; and said second end of said second portion of said first side wall and said fourth end of said fourth portion of said second side wall being spaced further apart than said first junction and said second junction.

13. The overload protection device of claim 12, wherein said elongated aperture portion of said disc shaft aperture comprises: a first interior angle between said first portion and said second portion of said first side wall being greater than 180 degrees; and a second interior angle between said third portion and said fourth portion of said second side wall being greater than 180 degrees.

14. The overload protection device of claim 13, wherein: said disc shaft has an elongated portion having a rounded rectangular shaped cross-section with opposing first and second shaft flat side walls.
15. The overload protection device of claim 14, wherein: when said handle spline is in said first position and said disc shaft is in said home angular position, said first portion of said first side wall of said disc shaft aperture extends away from said first shaft flat side wall of said disc shaft by said lost motion angle and said fourth portion of said second side wall of said disc shaft aperture extends away from said second shaft flat side wall of said disc shaft by said lost motion angle.
16. The overload protection device of claim 15, wherein: when said handle spline is in said first position and said disc shaft is in said home angular position, said second portion of said first side wall of said disc shaft aperture extends away from said first shaft flat side wall of said disc shaft by said lost motion angle and said third portion of said second side wall of said disc shaft aperture extends away from said second shaft flat side wall of said disc shaft by said lost motion angle.
17. The overload protection device of claim 16, wherein: when said handle spline is in said first position, said disc shaft is in said home angular position and rotationally decoupled from said disc shaft aperture of said handle spline, rotating said handle spline in said first rotational direction at least said lost motion angle frictionally engages said second portion of said first side wall and said third portion of said second side wall of said disc shaft aperture with said first shaft flat side wall and said second shaft flat side wall, respectively, of said disc shaft.
18. The overload protection device of claim 6, wherein: said stop flange is integrally formed with a spring bracket; and said spring bracket is fixedly coupled to said handle spline.
19. The overload protection device of claim 18, wherein: said handle inertia spring is operatively coupled between said spring bracket and the vehicle seat.
20. The overload protection device of claim 19, wherein: said handle spline has an outer profile; and said spring bracket has a main bracket portion, an alignment aperture passing through said main bracket portion, said stop flange projecting from said main bracket portion, and said alignment aperture having an inner profile configured to matingly engage with said outer profile of said handle spline.
21. The overload protection device of claim 20, wherein: said handle inertia spring is operatively coupled between said spring bracket and the vehicle seat, said handle inertia spring rotationally biasing said stop flange of said spring bracket towards abutment with said overload stop.
22. The overload protection device of claim 21, wherein: said recliner including a return spring operatively coupled between said recliner and said disc shaft, said return spring biasing said disc shaft towards said home angular position; and said handle spline being biased towards said first position by said handle spline being fixedly coupled to said spring bracket and said spring bracket being biased towards abutment with said overload stop by said handle inertia spring.
23. The overload protection device of claim 22, wherein when said recliner handle is unloaded by an externally-applied load: said return spring automatically rotates said disc shaft to said home angular position; said handle inertia spring automatically rotates said spring bracket to abutment with said overload stop; said handle spline automatically rotates to said first position by said spring bracket rotating to abutment with said overload stop; and said recliner handle automatically rotates to said home position by said handle spline rotating to said first position.
24. The overload protection device of claim 23, wherein: said spring bracket comprising a first flange and a second flange projecting from said main bracket portion with said first flange including a spring slot; and said handle inertia spring being a coil spring having a coiled portion with a central passageway extending through said coiled portion, a first spring end, and a second spring end, said first spring end configured to pass through said spring slot in said first flange, and said second spring end configured to pass through a spring retention hole in the vehicle seat, and said handle spline and said second flange passing through said central passageway.
25. The overload protection device of claim 24, wherein: said outer profile of said handle spline

including a first alignment feature configured to matingly engage with a second alignment feature of said inner profile of said alignment aperture of said spring bracket.

26. The overload protection device of claim 1, wherein: said stop flange is integrally formed with said handle spline.